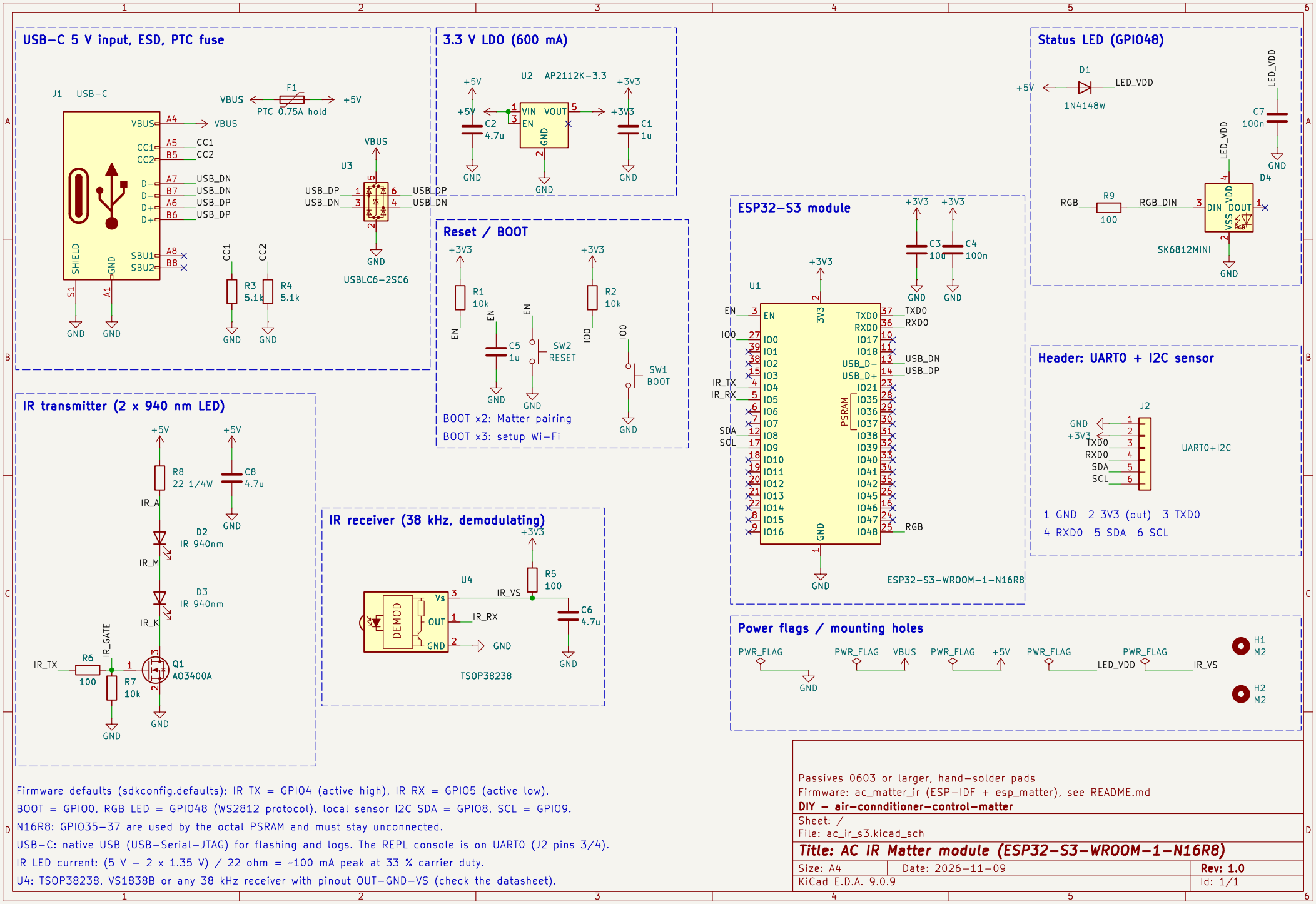


Firmware defaults (sdkconfig.defaults): IR TX = GPIO4 (active high), IR RX = GPIO5 (active low),
BOOT = GPIO0, RGB LED = GPIO48 (WS2812 protocol), local sensor I2C SDA = GPIO8, SCL = GPIO9.
N16R8: GPIO35–37 are used by the octal PSRAM and must stay unconnected.
USB-C: native USB (USB-Serial-JTAG) for flashing and logs. The REPL console is on UART0 (J2 pins 3/4).
IR LED current: (5 V − 2 × 1.35 V) / 22 ohm = ~100 mA peak at 33 % carrier duty.
U4: TSOP38238, VS1838B or any 38 kHz receiver with pinout OUT-GND-VS (check the datasheet).

Passives 0603 or larger, hand-solder pads
Firmware: ac_matter_ir (ESP-IDF + esp_matter), see README.md
DIY – air-connditioner-control-matter

Sheet: /
File: ac_ir_s3.kicad_sch

Title: AC IR Matter module (ESP32-S3-WROOM-1-N16R8)

Size: A4	Date: 2026-11-09	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1